Two exact identities for the magnetic energy, a ground-state sum rule, and why pointwise Gibbs domination fails as Agmon does

The magnetic energy of the Kogut–Susskind Hamiltonian satisfies two exact relations on the configuration space $G^{\mathcal{E}}$:

$$\Delta V = 4C_2(N|\mathcal{P}| - V), \qquad |\nabla V|^2 \;\le\; 16\,V \quad (SU(2)),$$

the first because $V - N|\mathcal{P}|$ is an eigenfunction of the Laplace–Beltrami operator with eigenvalue $-4C_2$, the second from the completeness relation together with $\sin^2\theta = (1-\cos\theta)(1+\cos\theta)$. They give an exact sum rule for the ground state,

$$A'\int (V - N|\mathcal{P}|)\,|\nabla\Omega|^2 d\mu + A'(2C_2 - K)\langle V - N|\mathcal{P}|\rangle + B'\,\mathrm{Var}_\Omega(V) = 0, \qquad K = \int |\nabla\Omega|^2,$$

and they make the comparison-principle attempt at a pointwise bound explicit: with $\varphi = e^{-\lambda V/2}$ the operator $L = -A'\Delta + (B'V - e_0)$ satisfies $L\varphi \ge 0$ precisely above a threshold $V_* \simeq N|\mathcal{P}|$, so the maximum principle bounds Ω only above the extensive Haar mean. **The pointwise Gibbs domination therefore fails for the same reason as the Agmon bound**: the ground-state energy e_0 is extensive and enters every comparison, so a local excess is never in the region where the comparison bites. The obstruction is that the Hamiltonian ground state is characterized by a global variational principle while the Euclidean weight e^{-S_w} is local by construction, and no manipulation of the eigenvalue equation alone repairs that. Constants explicit; the two identities and the sum rule are exact; nothing promoted.

1. The Laplacian of the magnetic energy

Proposition 1. On $\mathcal{M} = G^{\mathcal{E}}$ with the product bi-invariant metric,

$$\Delta V = 4C_2(R_{\mathrm{f}})(N|\mathcal{P}| - V),$$

where $C_2(R_{\mathrm{f}})$ is the quadratic Casimir of the defining representation. Equivalently $W = V - N|\mathcal{P}|$ satisfies $\Delta W = -4C_2 W$: the magnetic energy, shifted by its Haar mean, is an eigenfunction of the Laplacian.

Proof. Fix a plaquette p and one of its four links ℓ. Writing $\mathrm{tr}\,U_p = \sum_{ab}(U_\ell)_{ab}M_{ba}$ with M the ordered product of the other three links, each matrix element of U_ℓ in the defining representation is an eigenfunction of Δ_ℓ with eigenvalue $-C_2(R_{\mathrm{f}})$, by Peter–Weyl. Hence $\Delta_\ell\,\mathrm{tr}\,U_p = -C_2\,\mathrm{tr}\,U_p$, and summing over the four links of p, $\Delta\,\mathrm{Re\,tr}\,U_p = -4C_2\,\mathrm{Re\,tr}\,U_p$. Summing over plaquettes, $\Delta V = \Delta\sum_p(N - \mathrm{Re\,tr}\,U_p) = 4C_2\sum_p \mathrm{Re\,tr}\,U_p = 4C_2(N|\mathcal{P}| - V)$. □

2. The gradient of the magnetic energy

Proposition 2. For $SU(2)$, $|\nabla V|^2 \le 16\,V$ pointwise on $\mathcal{M}$. For general $SU(N)$, $|\nabla V|^2 \le 16\,N\,V$.

Proof. For one link of one plaquette, the completeness relation of the upper-bound note Corollary 2 gives

$$\sum_a |X_a^\ell\,\mathrm{tr}\,U_p|^2 = \tfrac{1}{2}\Big[\,\mathrm{tr}(MM^\dagger) - \tfrac{1}{N}|\,\mathrm{tr}\,M|^2\Big] = \tfrac{1}{2}\Big[N - \tfrac{1}{N}|\,\mathrm{tr}\,U_p|^2\Big].$$

For $SU(2)$, writing $\mathrm{tr}\,U_p = 2\cos\theta$, this is $1 - \cos^2\theta = \sin^2\theta = (1-\cos\theta)(1+\cos\theta)$, while $V_p = 2 - 2\cos\theta$, so

$$|\nabla_\ell\,\mathrm{Re\,tr}\,U_p|^2 \le \sin^2\theta = \frac{V_p}{2}\Big(2 - \frac{V_p}{2}\Big) \le V_p.$$

Each link lies in four plaquettes, so by Cauchy–Schwarz $|\nabla_\ell V|^2 \le 4\sum_{p\ni\ell} V_p$, and summing over links, with four links per plaquette, $|\nabla V|^2 \le 4\sum_\ell \sum_{p\ni\ell} V_p = 16\sum_p V_p = 16V$. The general case replaces the bound $\sin^2\theta \le V_p$ by $\frac{1}{2}[N - |\operatorname{tr} U_p|^2/N] \le NV_p$, which follows from $|\operatorname{tr} U_p| \ge N - V_p$ when $V_p \le N$ and from the trivial bound otherwise. □

Both identities vanish at $U_p = 1$: a configuration with no magnetic energy has no magnetic gradient, as it must.

3. An exact ground-state sum rule

Proposition 3. With $W = V - N|\mathcal{P}|$, $K = \int |\nabla\Omega|^2 d\mu$, $\langle\cdot\rangle = \langle\Omega, \cdot\,\Omega\rangle$ and $\operatorname{Var}_\Omega(V) = \langle V^2\rangle - \langle V\rangle^2$,

$$A'\int W|\nabla\Omega|^2 d\mu + A'(2C_2 - K)\langle W\rangle + B'\operatorname{Var}_\Omega(V) = 0.$$

Proof. Multiply $-A'\Delta\Omega + (B'V - e_0)\Omega = 0$ by $W\Omega$ and integrate. Since

$$\int W\Omega\,\Delta\Omega = -\int W|\nabla\Omega|^2 - \int \Omega\,\nabla W\cdot\nabla\Omega,$$

$$\int \Omega\,\nabla W\cdot\nabla\Omega = -\tfrac{1}{2}\int (\Delta W)\Omega^2 = 2C_2\langle W\rangle,$$

by Proposition 1, so $\int W\Omega\Delta\Omega = -\int W|\nabla\Omega|^2 - 2C_2\langle W\rangle$ and the equation becomes $A'\int W|\nabla\Omega|^2 + 2A'C_2\langle W\rangle + B'\langle VW\rangle - e_0\langle W\rangle = 0$. Now $e_0 = A'K + B'\langle V\rangle = A'K + B'(\langle W\rangle + N|\mathcal{P}|)$ and $\langle VW\rangle = \langle W^2\rangle + N|\mathcal{P}|\langle W\rangle$, so the last two terms combine into $-A'K\langle W\rangle - B'\langle W\rangle^2 + B'\langle W^2\rangle$. □

The sum rule ties the magnetic variance of the ground state to its kinetic energy and to the correlation between magnetic energy and kinetic density. At $g \to \infty$ every term vanishes with B', and at finite coupling it is one exact constraint on the ground-state measure.

4. Pointwise Gibbs domination and where it stops

Try $\varphi = e^{-\lambda V/2}$, $\lambda > 0$, as a comparison function for $L = -A'\Delta + (B'V - e_0)$. Using $\Delta\varphi = \varphi[\frac{\lambda^2}{4}|\nabla V|^2 - \frac{\lambda}{2}\Delta V]$ and Propositions 1–2,

$$\frac{L\varphi}{\varphi} = B'V - e_0 - A'\frac{\lambda^2}{4}|\nabla V|^2 + A'\frac{\lambda}{2}\Delta V \;\ge\; \Big[B' - 4A'\lambda^2 - 2A'\lambda C_2\Big]V + \Big[2A'\lambda C_2 N|\mathcal{P}| - e_0\Big],$$

for $SU(2)$, using $|\nabla V|^2 \le 16V$ and Proposition 1.

Proposition 4. If λ satisfies $B' > 4A'\lambda^2 + 2A'\lambda C_2$, then $L\varphi \ge 0$ on $\{V \ge V_*\}$ with

$$V_* = \frac{(e_0 - 2A'\lambda C_2 N|\mathcal{P}|)_+}{B' - 4A'\lambda^2 - 2A'\lambda C_2},$$

and the maximum principle gives

$$\Omega(U) \;\le\; \Big(\max_{\{V<V_*\}}\frac{\Omega}{\varphi}\Big)\, e^{-\lambda V(U)/2} \qquad \text{for all } U.$$

Proof. Set $u = \Omega/\varphi$. From $L\Omega = 0$, $0 = -A'\varphi\Delta u - 2A'\nabla u\cdot\nabla\varphi + u\,L\varphi$. At an interior maximum of u one has $\nabla u = 0$ and $\Delta u \le 0$, so the first two terms are ≥ 0 and $u\,L\varphi \le 0$; since $u > 0$, the maximum lies in $\{L\varphi \le 0\} \subset \{V < V_*\}$. □

Where it stops. With $e_0 \le B'N|\mathcal{P}|$, the threshold obeys

$$V_* \;\le\; N|\mathcal{P}|\,\frac{B' - 2A'\lambda C_2}{B' - 4A'\lambda^2 - 2A'\lambda C_2} \;\xrightarrow[\lambda\to 0]{}\; N|\mathcal{P}|,$$

so the comparison function controls Ω only relative to its values on the set where V is below the **extensive** Haar mean. The bound that results, $\Omega \le C\,e^{-\lambda(V-V_*)/2}$ with C determined on $\{V < V_*\}$, is again a statement about the global excess. A local excess of n plaquettes leaves V far below $V_* \simeq N|\mathcal{P}|$, so the bound says nothing about it, exactly as in the global/local note.

5. The structural reason, stated once

Both attempts, Agmon and comparison, use the eigenvalue equation $-A'\Delta\Omega+(B'V-e_0)\Omega=0$, in which e_0 is the **total** ground-state energy and is extensive. Any inequality derived from it compares the total potential with the total energy, so it distinguishes configurations only by their global excess. The Euclidean weight $e^{-S_w}=\prod_p e^{-(2/g_E^2)V_p}$ factorizes over plaquettes and therefore separates configurations locally by construction, with no reference to a total energy.

This is the precise content of the observation in the typical-field note: the Hamiltonian formulation has a state and a global eigenvalue equation where the Euclidean formulation has a local weight. The three notes since then have tried the two standard ways of extracting a local statement from a global equation, and both stop at the same place. A local statement would need an input of a different kind, for instance a proof that the ground state is a Gibbs state for a local potential, or the transfer-matrix identification of Ω^2 with a Euclidean measure on a time slice, which is the route the constructive programme takes.

6. Consequence for STATE

The question named in the global/local note §4 is answered in the negative by Proposition 4: pointwise Gibbs domination of the Kogut–Susskind ground state does not follow from the eigenvalue equation, for the same extensivity reason as the Agmon bound. What the attempt produced instead is worth keeping: the exact relations $\Delta V=4C_2(N|\mathcal{P}|-V)$ and $|\nabla V|^2\le 16V$, and the exact ground-state sum rule of Proposition 3, which constrains the magnetic variance in terms of kinetic quantities. The line of attack that remains is the transfer-matrix identification of the ground-state measure with a Euclidean measure, carried out in the transfer note: $|\Omega|^2 d\mu$ is the time-slice marginal of the four-dimensional Wilson measure, so every local Euclidean estimate transfers, and the local large-field bound becomes available in its standard scale-invariant form $e^{-c\eta^2/g^2}$.